

## **NATHAN PARK**

Berkeley, CA | (201) 906-6983 | nathan\_park@berkeley.edu | linkedin.com/in/parknathan1

### **PROFESSIONAL SUMMARY**

Mechanical engineer and UC Berkeley/UCSF Master of Translational Medicine student focused on medical-device and wearable product development. Experience translating clinical needs into requirements, prototypes, physiological-sensing systems, and bench/V&V evidence across R&D and GMP manufacturing.

### **EDUCATION**

**University of California, Berkeley & UC San Francisco** | Berkeley / San Francisco, CA | Jun 2027

Master of Translational Medicine, Medical Devices Track – Joint Program

**Vanderbilt University** | Nashville, TN | May 2026

Bachelor of Engineering in Mechanical Engineering, Minor in Engineering Management | GPA: 3.6/4.0

### **TECHNICAL SKILLS**

SOLIDWORKS, Python, C/C++, MATLAB, LabVIEW, Arduino, ESP32, Ansys, SLA/SLS/FDM, materials selection/evaluation, rapid prototyping, wearable sensor integration, and textile attachment design

### **PROFESSIONAL EXPERIENCE**

**Founder / Lead Wearable R&D** | **Astro Flexion** | Jul 2024 - Sep 2026

- Founded an NSF-supported wearable medical device venture, translating 100+ Voice of Customer (VoC) interviews into requirements for rehabilitation and human-performance monitoring.
- Designed and iterated textile MMG wearables; sensor integration, data acquisition, and signal processing, reducing signal noise by 38% with adaptive filtering and sensor-placement optimization.

**Technical Services / Manufacturing Science Validation Intern** | **Eli Lilly** | Concord, NC | May - Aug 2025

- Developed and deployed cross-functional equipment-handoff application used by 60+ stakeholders, eliminating manual handoff-status verification and streamlining interdependent site activities.
- Built and deployed Gantt dashboards for site V&V actions, identifying 107 recoverable schedule days.
- Developed an SOP and operator decision tool that standardized parts-washer inspection decisions, reducing equipment-driven downtime by 43% and supporting FDA/GMP-compliant execution.

### **RESEARCH AND SELECTED PROJECTS**

**Project Lead and Project Manager** | **Piccolo Medical / Spectrum Vascular** | Sep 2026 - Present

- Leading a four-person MTM capstone for Piccolo Medical / Spectrum Vascular, defining ECMO cannula guidance use cases, early requirements, risks, and a benchtop feasibility plan.

**Research Engineer** | **Vanderbilt IoMT Lab / VISE Group, Dr. Weimer** | Nov 2025 - Jun 2026

- Translated VUMC clinician needs into engineering requirements, test objectives, and prototype architectures for an ICU pulse oximetry patch and other clinical devices.
- Led development and testing of PHOENIX, a low-cost passive thermochemical blood warmer for austere care that heats 500 g of blood analog from 6°C to 30°C in 10 minutes.
- Built instrumented test setups and analyzed thermal trials across material and configuration variants, informing system thermal architecture and leading ongoing manuscript preparation.

**Senior Design Project** | **Circadian Rhythm Wearable for NIDDK** | Aug 2025 - May 2026

- Integrated light, PPG, temperature, and IMU sensors into an ESP32-based textile wearable; configured sensor placement, I<sup>2</sup>C data acquisition, and designed and executed human-use trials.
- Defined system requirements and architecture, authored V&V protocols/reports, and benchmarked sensor outputs against Apple Watch and ActiGraph wearables, demonstrating >80% cross-device agreement for PPG-derived heart rate, IMU, and temperature measurements.

**Undergraduate Research Assistant** | **Vanderbilt Miniature Robotics / Dong Lab** | Aug - Dec 2023

- Co-authored a 2024 PNAS paper on an artificial-cilia airway monitoring system; designed test fixtures and conducted magnetic-actuation experiments, device characterization, and data analysis.

### **CERTIFICATIONS & LEADERSHIP**

**Certifications:** CAPM, SOLIDWORKS Mechanical Design Associate, Linear Stress Analysis (Ansys)

**Leadership:** President & Board Member, Vanderbilt Relay for Life – Led 80-member board; chapter raised \$200k+ for the American Cancer Society over four years and ranked #9 nationally.